GR6082_PSet01

Problem Set 1: Correlations, Fermi Gas, and Electron Transport

GR6082 Condensed Matter Physics I — Spring 2026

Due: Sunday, February 22, 2026, 11:59 PM

Topics: Pair correlations, structure factor, Fermi gas, Sommerfeld model, Pauli paramagnetism, Drude model, Hall effect, Wiedemann-Franz law

---

Using AI for Computational Problems

This course includes computational problems designed to build physics intuition that equations alone cannot provide. You are encouraged to use AI assistants (Claude, ChatGPT, etc.) to help write code.

The rules:

1. **You write the prompt.** We will not give you a ready-made prompt. Translating a physics question into a computational task *is* the skill. A prompt that produces correct, insightful output requires understanding the physics first.
2. **You own the output.** AI-generated code often contains subtle errors — wrong units, missing factors of 2π, incorrect limiting behavior, unphysical parameter choices. You learn also by fixing these errors.
3. **Verify against known limits.** Every result must be checked against analytical results you derived (or can derive). If your code gives $\sigma(\omega \to 0) \neq \sigma_0$, something is wrong. State explicitly which limits you checked.
4. **All written answers must be handwritten** (scanned PDF, combined), including interpretations of computational results. Submit a **separate Jupyter notebook** with code and figures only. The notebook generates the plots; your handwritten solutions explain the physics.

Problem 1: Pair Correlation in a 1D Gas (20 points)

Consider a classical 1D gas of hard rods of length σ, with average density $n = N/L$.

(a) Sketch the pair correlation function $g(r)$ for this system. What is $g(r)$ for $r < \sigma$? What happens as $r \to \infty$?

(b) The static structure factor is related to the pair correlation function by:

$$S(q) = 1 + n \int_{-\infty}^{\infty} [g(r) - 1] e^{iqr} dr$$

What is $S(q = 0)$ for this system? Interpret this result in terms of density fluctuations.

(c) For a dilute gas ($n\sigma \ll 1$), approximate $g(r) \approx \Theta(r - \sigma)$ where Θ is the Heaviside step function. Compute $S(q)$ and sketch it.

Problem 2: The Fermi Gas at Zero Temperature (20 points)

The Fermi gas exerts pressure even at $T = 0$ due to the Pauli exclusion principle (degeneracy pressure). This is what supports white dwarf stars against gravitational collapse.

(a) Calculate the total kinetic energy E_{tot} of a 3D free electron gas at $T = 0$ by integrating over the Fermi sphere:

$$E_{tot} = 2 \int_{|\boldsymbol{k}|<k_F} \frac{d^3k}{(2\pi)^3} \frac{\hbar^2 k^2}{2m}$$

Periodic boundary conditions?

E_tot - energy per unit volume

Show that the average energy per electron is $\langle \varepsilon \rangle = \frac{3}{5}\varepsilon_F$.

(b) Using the thermodynamic relation $P = -(\partial E_{tot}/\partial V)_N$ at fixed N, show that:

$$P = \frac{2}{3}\frac{E_{tot}}{V}$$

Hint: Express E_{tot} in terms of N and V, then differentiate. In b E_tot -> total energy

(c) Calculate this degeneracy pressure (in Pascals) for a typical metal with $n = 10^{29}$ m^{-3}. Compare to atmospheric pressure ($\sim 10^5$ Pa). Why doesn't this enormous pressure cause metals to explode?

(d) In a white dwarf star, electron degeneracy pressure supports the star against gravity. Estimate the electron density needed to support a star of mass $M \sim 10^{30}$ kg and radius $R \sim 10^7$ m. Compare to metallic densities.

Problem 3: Sommerfeld Expansion (20 points)

At low temperatures, the Fermi-Dirac distribution can be expanded using the Sommerfeld formula:

$$\int_0^\infty H(\varepsilon) f(\varepsilon) d\varepsilon \approx \int_0^\mu H(\varepsilon) d\varepsilon + \frac{\pi^2}{6}(k_B T)^2 H'(\mu) + O(T^4)$$

f(eps) - fermi dirac

H(eps) - any function

(a) Use this to show that the chemical potential at low temperature is:

$$\mu(T) \approx \varepsilon_F \left[1 - \frac{\pi^2}{12}\left(\frac{k_B T}{\varepsilon_F}\right)^2\right]$$

for 3D electron gas?

Hint: Use the fact that at $T = 0$, $\mu = \varepsilon_F$, and expand around this.

(b) The total energy of the electron gas is $E = \int_0^\infty \varepsilon\, g(\varepsilon) f(\varepsilon) d\varepsilon$, where $g(\varepsilon) \propto \varepsilon^{1/2}$ is the 3D density of states. Use the Sommerfeld expansion to show: At T -> 0

$$E(T) \approx E_0 + \frac{\pi^2}{6} g(\varepsilon_F)(k_B T)^2$$

(c) From this, derive the electronic heat capacity $C_V = \gamma T$ and express the Sommerfeld coefficient γ in terms of $g(\varepsilon_F)$.

Problem 4: Pauli Paramagnetism (20 points)

A free electron gas in a magnetic field $\boldsymbol{B} = B\hat{z}$ is one of the simplest problems in condensed matter — yet the result is strikingly different from what classical physics predicts.

(a) Classical expectation. Treat each electron as an independent magnetic moment μ_B. Using Curie's law for non-interacting paramagnetic moments, the classical susceptibility is:

$$\chi_{\text{Curie}} = \frac{\mu_0 n \mu_B^2}{k_B T}$$

Evaluate this for copper ($n = 8.5 \times 10^{28}$ m^{-3}) at room temperature. Is it temperature-dependent? How large is it?

(b) Quantum treatment. In a magnetic field, the single-particle energies split:

$$\varepsilon_\uparrow(\boldsymbol{k}) = \frac{\hbar^2 k^2}{2m} + \mu_B B, \qquad \varepsilon_\downarrow(\boldsymbol{k}) = \frac{\hbar^2 k^2}{2m} - \mu_B B$$

At $T = 0$, both spin species fill up to the same chemical potential μ. By computing $N_\uparrow$ and $N_\downarrow$ for small B (i.e., $\mu_B B \ll \varepsilon_F$), show that the magnetization is:

$$M = \mu_B^2 \, g(\varepsilon_F) \, B$$

and hence the Pauli susceptibility is:

$$\chi_P = \mu_0 \mu_B^2 \, g(\varepsilon_F)$$

Hint: The number of spin-↑ electrons is $N_\uparrow = \frac{1}{2} \int_0^{\mu - \mu_B B} g(\varepsilon) \, d\varepsilon$. Expand for small B.

(c) Show that:

$$\frac{\chi_P}{\chi_{\text{Curie}}} \sim \frac{k_B T}{\varepsilon_F} \ll 1$$

Evaluate numerically for copper at $T = 300$ K ($\varepsilon_F \approx 7$ eV). Why is Pauli paramagnetism so much weaker than the classical prediction? What is the physical picture — which electrons can actually respond to the field?

(d) Use the Sommerfeld expansion to show that the leading temperature correction to the susceptibility is:

$$\chi(T) = \chi_P \left[1 - \frac{\pi^2}{12} \left(\frac{k_B T}{\varepsilon_F} \right)^2 + \cdots \right]$$

Is the Pauli susceptibility essentially temperature-independent for metals at room temperature?

Problem 5: Drude Optical Response (30 points)

Analytical + Computational

Part I: Analytical (15 points)

Consider the Drude model for electrons in a metal with scattering time τ.

(a) Starting from the equation of motion $m\dot{\boldsymbol{v}} = -e\boldsymbol{E} - m\boldsymbol{v}/\tau$, derive the AC conductivity:

$$\sigma(\omega) = \frac{\sigma_0}{1 - i\omega\tau}$$

where $\sigma_0 = ne^2\tau/m$.

(b) Show that the real and imaginary parts of the conductivity are:

$$\sigma_1(\omega) = \frac{\sigma_0}{1 + \omega^2\tau^2}, \quad \sigma_2(\omega) = \frac{\sigma_0 \omega\tau}{1 + \omega^2\tau^2}$$

1/tau << omega_p

(c) The dielectric function is related to the conductivity by $\epsilon(\omega) = 1 + i\sigma(\omega)/(\epsilon_0\omega)$. Derive the plasma frequency $\omega_p = \sqrt{ne^2/(\epsilon_0 m)}$ at which $\epsilon(\omega) = 0$. For copper ($n = 8.5 \times 10^{28}$ m^{-3}), calculate ω_p and the corresponding wavelength. Is copper transparent to visible light?

Part II: Computational (15 points)

Now visualize the full optical response. Use copper parameters ($n = 8.5 \times 10^{28}$ m^{-3}, $\tau = 2.5 \times 10^{-14}$ s).

(d) Create a multi-panel figure showing the frequency dependence of: (i) the complex conductivity $\sigma_1(\omega)$ and $\sigma_2(\omega)$; (ii) the phase angle $\phi(\omega) = \arctan(\sigma_2/\sigma_1)$ between $\boldsymbol{j}$ and $\boldsymbol{E}$; (iii) the dissipated power $P(\omega)/P(0) = \sigma_1(\omega)/\sigma_0$; (iv) the complex dielectric function $\epsilon_1(\omega)$ and $\epsilon_2(\omega)$; and (v) the reflectivity $R(\omega) = \left|\frac{\sqrt{\epsilon(\omega)}-1}{\sqrt{\epsilon(\omega)}+1}\right|^2$. Use a log scale for frequency (in eV). Mark the key energy scales: $\hbar/\tau$ and $\hbar\omega_p$.

(e) From your plots, answer:

- At what frequency does the response cross over from resistive ($\sigma_1 > \sigma_2$) to reactive ($\sigma_2 > \sigma_1$)? What is the physical meaning of each regime?
- What is the phase ϕ between $\boldsymbol{j}$ and $\boldsymbol{E}$ in each regime? What does $\phi = \pi/4$ correspond to physically?
- At what frequency has the dissipated power dropped to half its DC value? Relate this to a characteristic timescale.
- Where does the reflectivity drop sharply? How does this explain why metals are shiny? Where does the visible spectrum sit relative to ω_p for copper?

(f) The optical conductivity satisfies the f-sum rule: $\int_0^\infty \sigma_1(\omega)\, d\omega = \frac{\pi n e^2}{2m}$. Verify this numerically for your Drude $\sigma_1(\omega)$. What fraction of the spectral weight lies below $1/\tau$?

Problem 6: Hall Effect and Magnetoresistance (20 points)

Consider a conductor with carrier density n in a static magnetic field $\boldsymbol{B} = B\hat{z}$ and an electric field $\boldsymbol{E}$ in the xy-plane.

(a) Using the steady-state Drude equation of motion ($d\boldsymbol{p}/dt = 0$):

$$0 = -e(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) - \frac{m\boldsymbol{v}}{\tau}$$

derive the components of the resistivity tensor ρ_{ij} defined by $E_i = \sum_j \rho_{ij} j_j$.

R_H = rho_yx/B

(b) Show that the Hall coefficient $\boxed{R_H \equiv \rho_{xy}/B}$ is given by $R_H = -1/(ne)$. What is the physical origin of the Hall voltage?

(c) The Drude model predicts that the longitudinal resistivity ρ_{xx} is independent of B (no classical magnetoresistance). Verify this from your derivation in part (a). Why does this result fail in real materials?

(d) In a Hall bar geometry, $j_y = 0$ by construction. Express E_y/E_x in terms of $\omega_c\tau$ where $\omega_c = eB/m$ is the cyclotron frequency. What happens in the limits $\omega_c\tau \ll 1$ and $\omega_c\tau \gg 1$?

Problem 7: Thermal Conductivity and Wiedemann-Franz (20 points)

(a) In the Drude model, the thermal conductivity is given by:

$$\kappa = \frac{1}{3} C_V v^2 \tau$$

where C_V is the heat capacity per unit volume and v is a characteristic velocity. Using the classical result $C_V = \frac{3}{2} n k_B$ and $\frac{1}{2} m v^2 = \frac{3}{2} k_B T$, show that the Lorenz number is:

$$L = \frac{\kappa}{\sigma T} = \frac{3}{2}\left(\frac{k_B}{e}\right)^2$$

sigma -> sigma_0

(b) In the Sommerfeld model, only electrons near ε_F contribute. Repeat the calculation using $C_V = \gamma T$ (with $\gamma = \frac{\pi^2}{2} n k_B^2/\varepsilon_F$) and $v = v_F$. Show that:

$$L = \frac{\pi^2}{3}\left(\frac{k_B}{e}\right)^2$$

(c) Compare these predictions with the experimental value $L \approx 2.4 \times 10^{-8}$ W·Ω/K^2. Which model gives the correct answer?

(d) The Drude model uses classical statistics, which:

- Overestimates the heat capacity: $C_V^{Drude} = \frac{3}{2} n k_B$ vs. $C_V^{Sommerfeld} \propto T$
- Underestimates the velocity: $\langle v^2 \rangle^{Drude} = 3k_B T/m$ vs. v_F^2 (temperature-independent)

Explain qualitatively why these two errors approximately cancel, allowing Drude to obtain a Lorenz number within a factor of 2 of the correct value. This "Wiedemann-Franz miracle" delayed the discovery of quantum statistics by decades!

Problem 8: Gapped vs Gapless (30 points)

Analytical + Computational

One of the most important distinctions in condensed matter physics is between **gapped** and **gapless** systems. The presence or absence of a spectral gap in the density of states $g(\varepsilon)$ controls the low-temperature behavior of essentially every thermodynamic quantity.

Part I: Analytical (10 points)

(a) Show that for a gapless system with $g(\varepsilon) \propto \varepsilon^\alpha$ near $\varepsilon = 0$:

Of fermions

$$C_{el}(T) \propto T^{1+\alpha}$$

Verify the familiar cases: constant DOS ($\alpha = 0$) gives $C \propto T$ (metal); linear DOS ($\alpha = 1$) gives $C \propto T^2$ (2D Dirac fermion); quadratic DOS ($\alpha = 2$) gives $C \propto T^3$ (3D acoustic phonons).

(b) Show that for a gapped system with gap Δ and roughly constant DOS above the gap, the low-temperature specific heat goes as:

$$C_{el}(T) \propto \left(\frac{\Delta}{k_B T}\right)^2 e^{-\Delta/k_B T}$$

Hint: The key contribution comes from thermally activated excitations across the gap.

Part II: Computational (20 points)

(c) Before analyzing data, build intuition. For a gapped system with $\Delta = 5$ K, plot the occupation probability $f(\Delta, T) = 1/(e^{\Delta/k_B T} + 1)$ as a function of temperature. At what T is the probability of occupying a state just above the gap equal to 1%? To 10%? Then overlay the occupied density of states $g(\varepsilon) \times f(\varepsilon, T)$ for both a gapless system ($g =$ const) and a gapped system ($g = 0$ for $\varepsilon < \Delta$) at three temperatures ($T = \Delta/10$, $\Delta/2$, 2Δ). Explain in your own words why the gapped system's specific heat is exponentially suppressed at low T.

You receive electronic specific heat data (phonon contribution subtracted) from four unknown materials:

T (K)	System A	System B	System C	System D
0.5	0.33	0.04	< 0.01	< 0.01
1.0	0.73	0.16	< 0.01	0.19

T (K)	System A	System B	System C	System D
1.5	1.08	0.35	0.01	0.98
2.0	1.36	0.58	0.03	2.55
3.0	2.17	1.41	0.32	6.0
4.0	2.76	2.42	1.9	8.0
5.0	3.62	3.88	5.4	10.0
7.0	5.12	7.55	14.5	14.0
10.0	6.79	15.6	21.8	20.0
15.0	10.8	32.8	18.4	30.0
20.0	14.4	62.1	13.3	40.0

(C_{el} in mJ/mol·K)

(d) For each system, make two diagnostic plots:

1. $\ln C_{el}$ vs $\ln T$ — a power law $C \propto T^n$ appears as a **straight line** with slope n
2. $\ln C_{el}$ vs $1/T$ — an activated form $C \propto e^{-\Delta/T}$ appears as a **straight line** with slope $-\Delta$

Determine which systems are gapless (power law) and which are gapped (exponential). For each power-law system, extract the exponent α. For each gapped system, extract Δ.

(e) Connect the extracted exponents to the density of states:

- What does $\alpha = 0$ imply about $g(\varepsilon)$? What kind of system has this?
- What does $\alpha = 1$ imply? Give a physical example.
- For the gapped system(s), what is the gap Δ in meV? In Kelvin?

(f) One of the four systems has a **phase transition** — it is gapless at high temperature but develops a gap at low temperature. Identify it from the data. Estimate the transition temperature T_c. What kind of phase transition could produce this behavior?

Deliverables: (1) Occupation probability and occupied-DOS visualization from (c). (2) Two diagnostic plots (log-log and Arrhenius) with all four systems overlaid. (3) Table of extracted exponents or gaps. (4) Identification of each system's physics.

Problem 9: The Strange Metal (20 points)

This problem connects to Problem 8: $CeAl_3$ is gapless but anomalous — the standard Sommerfeld model fails spectacularly.

Computational

You're studying a new intermetallic compound. Specific heat measurements give:

T (K)	C (mJ/mol·K)
2.0	3130
3.0	5200

T (K)	C (mJ/mol·K)
4.0	7350
5.0	9250
6.0	10900
8.0	13900
10.0	16900
15.0	24500
20.0	33000
30.0	52000
50.0	79000
100.0	96000
200.0	101000
300.0	103000

(a) Attempt to fit this data with the standard model: $C = \gamma T + \beta T^3$ at low T (below 10 K). Plot C/T vs T^2. What values do you extract for γ and Θ_D? Does the fit work? Is the intercept γ physically reasonable?

(b) The standard fit fails — the data curves strongly rather than being linear. Propose and computationally test at least two alternative models:

- Einstein modes (localized vibrations) in addition to Debye phonons?
- Non-Fermi liquid behavior: $C_{el} \propto T \ln(T_0/T)$ or $C_{el} \propto T^n$ with $n \neq 1$?
- Schottky anomaly from crystal-field-split levels?
- Nuclear contribution at very low T?

Which modification best describes the data across the full temperature range?

(c) The compound has formula $\mathrm{CeAl_3}$. Given your analysis, what can you infer about the electronic structure? Why might the standard Sommerfeld model fail?

(d) Heavy-fermion materials show γ values 100–1000$\times$ larger than normal metals. From your best fit, estimate the effective mass m^*/m_e. If this material becomes superconducting below 0.5 K, what unusual properties might you expect?

Deliverables: (1) C/T vs T^2 plot showing failure of standard fit. (2) Comparison of at least two alternative models with data. (3) Table of best-fit parameters. (4) Brief written answers for (c) and (d).

Feedback

This is Problem Set 1 — help me calibrate. After completing the assignment, briefly answer (here or anonymously to the TA):

1. **Difficulty:** Was this problem set too easy, about right, or too hard? Which problems were most/least challenging?
2. **Length:** Did the workload feel appropriate for one week?
3. **Computational problems:** Was the level of guidance sufficient? Would you prefer more structure (e.g., suggested approaches) or less? Are the computational parts adding to your understanding, or just busywork?
4. **AI usage:** How did you use AI tools? What worked well, what didn't?

Total: 200 points (140 analytical + 60 computational)